

SignalDesk — a news ranking engine that shows its work

A Telegram bot that reads 12 crypto and macro feeds, ranks every story with an auditable eight-factor score, and lets a language model write only what the algorithm already chose.

PROBLEM STATEMENT

Twelve newsfeeds produce a few hundred crypto and macro headlines a day — mostly listicles, price-prediction bait, and the same story told six times by six outlets. The signal is in there; the cost of finding it is forty minutes every morning, and the reader who most needs a fast multi-angle read of the market is the one who cannot spend it.

The obvious fix — hand the pile to a language model — fails in a specific way. **The model's selection is unauditable.** When it leads with the wrong story you cannot find out why, cannot reproduce it, and have nothing to correct. Forty minutes of reading has been traded for a black box.

SOLUTION OVERVIEW

Split the job at its natural seam. **An algorithm decides what matters** — deterministic, inspectable, covered by 148 offline tests. **A model writes it up:** it receives the already-ranked list and is told in its system prompt that selection is not its job — no re-ordering, no added stories, no number absent from the input. Three commands — /digest (the briefing, pushed at 08:30), /top (the live ranking with scores), /weights (five subject filters and a numbers-vs-analysis depth setting that visibly moves two weights). Between briefings it re-checks every 15 minutes and pushes anything over threshold.

- 0 • **hard filters**
clickbait, price ticks, off-topic macro — binary
- 1 • **dedupe**
TF-IDF cosine + rare-entity overlap, transitive closure
- 2 • **eight-factor score**
keyword ·185 recency ·185 source ·185 topicality ·130 numeric ·100 analysis ·100 corroboration ·075 asset ·040
- 3 • **adaptive top-N**
quality gate, diversity cap; separate crypto/macro pools
- 4 • **LLM write-up → Telegram**
08:30 digest, threshold alerts, live commands

/why 1 – "Curve exploited for \$62m"			
factor	raw	wt	adds
keyword	1.00	0.185	0.185
source_quality	0.95	0.185	0.176
recency	0.94	0.185	0.173
topicality	0.50	0.130	0.065
numeric	0.35	0.100	0.035
asset	0.50	0.040	0.020
analysis	0.00	0.100	0.000
source_count	0.00	0.075	0.000
TOTAL			0.654

USE OF AI

Built with Claude Code as a pair programmer throughout — scaffolding, scoring, every test in the suite. **In the product:** Moonshot Kimi K3 with xAI Grok 4.3 as fallback, via TokenRouter — cross-vendor on purpose, because two models from one provider are a retry, not a fallback. **Free, unauthenticated data:** 12 RSS feeds (CoinDesk, The Block, DL News, FT, Bloomberg, ECB Press and six more), the **Binance** ticker and the **alternative.me Fear & Greed** index. When every provider fails the bot emits template output from the top-ranked headlines — degraded, honest, still correctly ordered. The system prompt is what enforces that boundary:

"You will be given a numbered list of stories that has ALREADY been selected and ranked by a scoring engine. **Selection is not your job.** Do not re-order them... **Never invent a number.** Prices and dollar amounts may only appear if they appear in the input... The headline **MUST** be about story 1 — choosing a different story is a selection decision."

IMPACT & VALUE

Forty minutes of triage becomes a thirty-second read, and — unlike a summariser — the output can be challenged: /why n answers "why is this at the top" with arithmetic, so a wrong lead story traces to a factor weight, not a sampling temperature. One LLM call per digest and a free data layer make the next step viable: a **free subscription site** where readers pick their subjects and get the same push — free tier as the growth channel, not a loss leader.

REFLECTIONS

Making the score explain itself changed how I built the whole thing. /why began as a nice-to-have and became the debugging tool — **every real bug here was found by reading the factor table beside a ranking, and not one was caught by a test.** A `\b(exploit)\b` pattern could not match "exploited", so a \$62m hack scored 0.00 on the heaviest factor and ranked 7th of 8; after the fix, 1.00 and 2nd. Three later bugs shared one habit — a broad word added to a category, checked only against the headline it was meant to catch — fixed by bidirectional tests.

Given more time: the eight weights are **reasoned, not fitted** — every one defensible, none measured. Logging each ranking against which stories readers open would turn eight arguments into eight numbers.